

Experiment 8: Multi-Agent Task Allocation and Coordination

Aim

To implement a simple multi-agent task allocation system using Partial Global Planning (PGP) and Contract Net Protocol (CNP).

Exercises

1. Agent Setup

Create a Python-based system with 1 Manager Agent and at least 3 Delivery Agents. Add at least 6 delivery tasks with different costs/times for each agent.

2. PGP Coordination

Create local plans for the agents. Create a conflict where two agents use the same location/resource at the same time. Detect the conflict and modify one of the plans to resolve it.

3. CNP Task Allocation

Implement the CNP process:

Task Announcement -> Bidding -> Bid Comparison -> Agent Selection -> Task Assignment.

4. Multiple Task Allocation

Allocate all tasks automatically using the agents' bids. Display the final allocation as:

Task	Selected Agent	Cost/Time
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Also calculate the total cost/time.

5. Modification & Visualization

Change some task/agent costs and run the system again. Compare the results and plot Agent vs Total Assigned Cost/Time using Matplotlib.

Observation

Observe how PGP resolves conflicts, how CNP assigns tasks, and how changing the input affects the final task allocation.